DSC 40B

Theoretical Foundations II

Lecture 7 | Part 1

Time Complexity of Mergesort

Today in DSC 40B...

- Finish Lecture 06: the time complexity of mergesort, and using sorted structure.
  - These are covered by Quiz 03.
- Then: quickselect.

Last Time: Mergesort

```
def mergesort(arr):
    """Sort array in-place."""
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        mergesort(left)
        mergesort(right)
        merge(left, right, arr)
```





- Copying the two halves takes $\Theta(n)$.
- `merge` takes $\Theta(n)$.

The Recurrence

```
def mergesort(arr):
    """Sort array in-place."""
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        mergesort(left)
        mergesort(right)
        merge(left, right, arr)
```

$\Theta(n)$

$T(n/2)$

$T(n/2)$

$T(n) = \Theta(n) + T(n/2) + T(n/2)$

$= \Theta(n) + 2T(n/2)$

Solving the Recurrence

$T(n) = 2T(n/2) + cn$

$T(n/2) = 2T(n/4) + n/2$

$T(n) = 2T(n/2) + n \qquad k=1$

$= 2\left[2T(n/4) + n/2\right] + n$

$= 4T(n/4) + n + n$

$= 4T(n/4) + 2n \qquad k=2$

Solving the Recurrence

$T(n) = 2T(n/2) + \Theta(n)$

$T(n/4) = 2T(n/8) + n/4$

$T(n) = 4T(n/4) + 2n \quad k=2$

$= 4\left[2T(n/8) + n/4\right] + 2n$

$= 8T(n/8) + n + 2n$

$= 8T(n/8) + 3n \quad k=3$

Solving the Recurrence

$T(n) = 2T(n/2) + \Theta(n)$

$T(n) = 2^k T(n/2^k) + kn$

Solving: $\frac{n}{2^k} = 1$ for $k \Rightarrow k = \log_2 n$

$T(n) = 2^{\log_2 n} T(n/2^{\log_2 n}) + (\log_2 n)\, n$

$= n\, T(n/n) + n \log_2 n$

$= n\, T(1) + n \log_2 n = n + n \log_2 n =$

$T(1) = \Theta(1)$

$\Theta(n \log n)$

Optimality

- **Theorem**: Any (comparison) sorting algorithm's worst-case time complexity must be $\Omega(n \log n)$.
- Mergesort is **optimal**!

Be Careful!

- It is possible for a sorting algorithm to have a **best case** time complexity smaller than $n \log n$.
  - Insertion sort, for example.
- Mergesort has best case time complexity of $\Theta(n \log n)$.
- Mergesort is **sub-optimal** in this sense!

Be Careful!

- The $\Theta(n \log n)$ lower-bound is for **comparison sorting**.
- It is possible to sort in worst-case $\Theta(n)$ time without comparing.[1]

[1] Bucket sort, radix sort, etc.

What if?

- **Divide**: split the array into halves
- **Conquer**: sort each half **using selection sort**
- **Combine**: merge sorted halves together

Live Q&A

Exercise

What is the time complexity of the code below?

A) $\Theta(n)$ B) $\Theta(n \log n)$ C) $\Theta(n^2)$ D) $\Theta(n^2 \log n)$

$\left(\frac{n}{2}\right)^2 = \frac{n^2}{4} = \Theta(n^2)$

$T(n) = \Theta(n) + \Theta(n) + \Theta(n^2)$

```
def mergeselectionsort(arr):
    """Sort array in-place."""
    if len(arr) > 1:
        middle = math.floor(len(arr) / 2)
        left = arr[:middle]
        right = arr[middle:]
        selection_sort(left)
        selection_sort(right)
        merge(left, right, arr)
```

DSC 40B

Theoretical Foundations II

Lecture 7 | Part 2

Using Sorted Structure (Optional)

Sorted structure is useful

- Some problems become **much easier** if input is sorted.
  - For example, median, minimum, maximum.
- Sorting is useful as a **preprocessing** step.

Recall: The Movie Problem

- You're on a flight that will last *D* minutes.
- You want to pick two movies to watch.
- You want the total time of the two movies to be **as close as possible** to *D*.

The Movie Problem

- Brute force algorithm: $\Theta(n^2)$
- We can do better, if movie times are **sorted.**

Example

- Flight duration $D = 155$
- Movie times: $60, 80, 90, 120, 130$

	60	80	90	120	130
60					
80					
90					
120					
130					

Best pair:

The Algorithm

- Keep index of shortest and longest remaining.
- On every iteration, pair the shortest and longest.
- If this pair is too long, remove longest movie; otherwise remove shortest.
  - If times are **sorted**, finding new longest/shortest movie takes $\Theta(1)$ time!

60, 80, 90, 120, 130

The Algorithm

```
def optimize_entertainment(times, target):
    """assume times is sorted."""
    shortest = 0
    longest = len(times) - 1

    best_pair = (shortest, longest)
    best_objective = float('inf')

    for i in range(len(times) - 1):
        total_time = times[shortest] + times[longest]

        if abs(total_time - target) < best_objective:
            best_objective = abs(total_time - target)
            best_pair = (shortest, longest)

        if total_time == target:
            return (shortest, longest)
        elif total_time < target:
            shortest += 1
        else: # total_time > target
            longest -= 1

    return best_pair
```

Main Idea

Sorted structure allows you to rule out possibilities without explicitly checking them. But, it requires you to spend the time sorting first.

Tip: when designing an algorithm, think about sorting the input first.

DSC 40B

Theoretical Foundations II

Lecture 7 | Part 3

The Median and Order Statistics

The Median

- How fast can we find a **median** of n numbers?

Algorithms

- We have seen several ways of computing a median:
  - Alg. 1: Minimize absolute error, brute force.
  - Alg. 2: Use definition (half ≤, half ≥).
  - ...

Exercise

Using what we know so far, what approach for finding the median has the best **worst-case time complexity**?

$\Theta(n^2)$

Best so far...

- Sort the list with mergesort, return middle element.
- Time complexity: $\Theta(n \log n)$.

Is sorting necessary?

- Need to sort the whole list just to find middle?
- Seems like more work than necessary.

Today

- We'll design an algorithm which runs in $\Theta(n)$ expected time.
- Much more useful than just finding median...

Order Statistics

- The median is an example of an **order statistic**.

Definition

Given n numbers, the **kth order statistic** is the kth smallest number in the collection.

Example

```
[99, 42, -77, -12, 101]
```

- 1st order statistic: -77
- 2nd order statistic: -12
- 4th order statistic: 99

Exercise

Some special cases of order statistics go by different names. Can you think of some?

Special Cases

- **Minimum**: 1st order statistic.
- **Maximum**: nth order statistic.
- **Median**: $\lceil n/2 \rceil$th order statistic[2].
- p**th Percentile**: $\lceil \frac{p}{100} \cdot n \rceil$th order statistic.

[2] What if n is even?

Goal

- **Fast** algorithm for computing any order statistic.
- Interestingly, some seem easier than others.
- Our algorithm will find **any** order statistic in $\Theta(n)$ *expected* time.

Approach #1

- We can modify `selection_sort` to find the *k*th order statistic.
- Loop invariant: after *k*th iteration, first *k* elements are in final sorted order.

```
def selection_sort(arr):
    """In-place selection sort."""
    n = len(arr)
    if n <= 1:
        return
    for barrier_ix in range(n-1):
        # find index of min in arr[start:]
        min_ix = find_minimum(arr, start=barrier_ix)
        #swap
        arr[barrier_ix], arr[min_ix] = (
                arr[min_ix], arr[barrier_ix]
        )
```

```
def select_k(arr, k):
    """Find kth order statistic."""
    n = len(arr)
    if n <= 1:
        return
    for barrier_ix in range(k):
        # find index of min in arr[start:]
        min_ix = find_minimum(arr, start=barrier_ix)
        #swap
        arr[barrier_ix], arr[min_ix] = (
                arr[min_ix], arr[barrier_ix]
        )
    return arr[k-1]
```

Exercise

What are the best case and worst case time complexities of `select_k`?

Approach #1

- 1st order statistic: $\Theta(n)$.
- nth order statistic: $\Theta(n^2)$.
- Median: $\Theta(n^2)$.
- kth order statistic: $\Theta(kn)$.

$n \Rightarrow \frac{n}{2}$

$n^2 \Rightarrow \frac{n^2}{4} = \Theta(n^2)$

Exercise

Describe how to find any order statistic in $\Theta(n \log n)$ time.

Approach #2

- Sort with mergesort, return `arr[k-1]`
- $\Theta(n \log n)$ time. Could be better...

DSC 40B

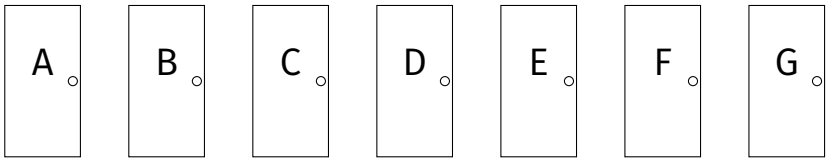
Theoretical Foundations II

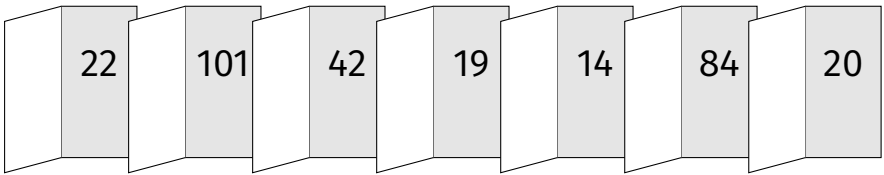
Lecture 7 | Part 4

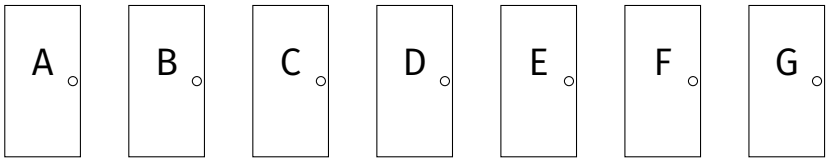
Quickselect

The Goal

- Given a collection of n numbers and an order, k.
- Find the kth smallest number in the collection.

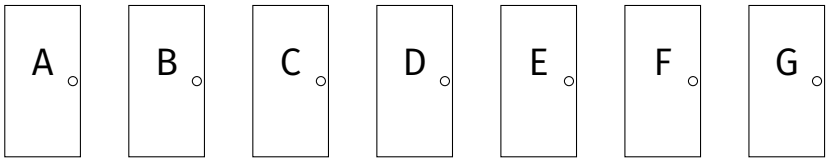

A
B
C
D
E
F
G

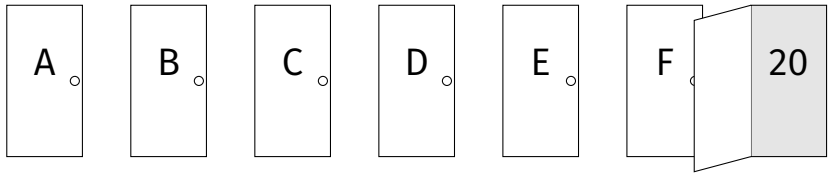

22
101
42
19
14
84
20

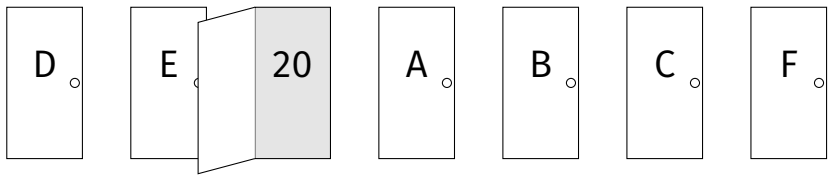

A
B
C
D
E
F
G

Game Show

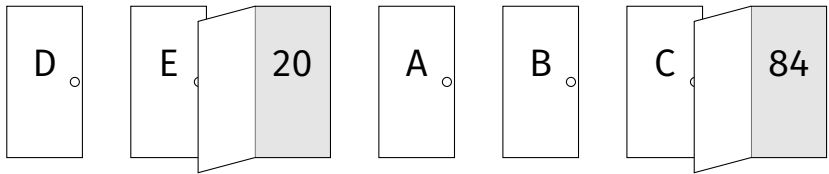
- **Goal**: tell the host the **largest** number.
- **Caution**: with every door opened, your money is reduced.
- **Twist**: After opening a door, the host tells you:
  - which doors are smaller.
  - which doors are larger.
  - they **partition** the doors into higher and lower by moving them.

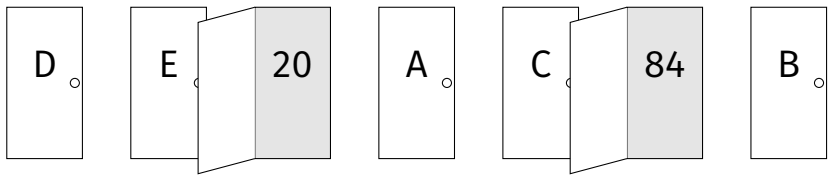

A
B
C
D
E
F
G


A
B
C
D
E
F
20

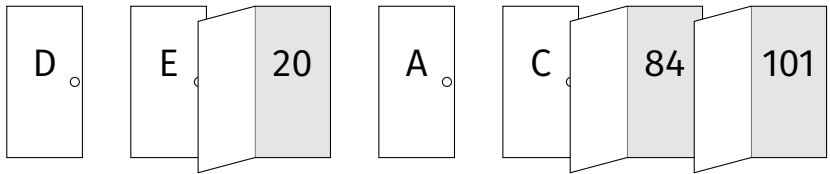


after partitioning


D
E
20
A
B
C
84



after partitioning


D
E
20
A
C
84
101

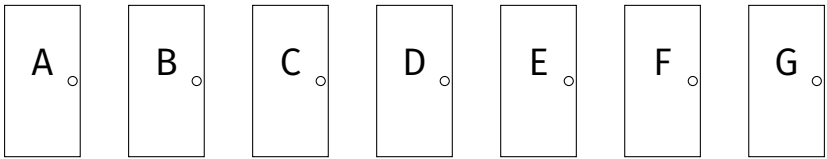
Main Idea

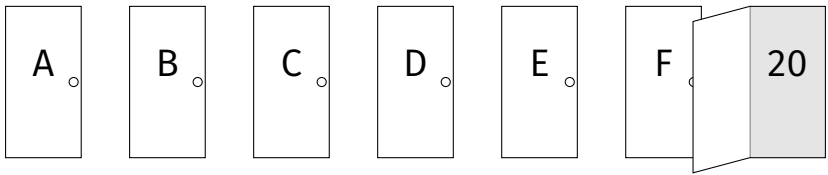
After partitioning, the just-opened door is in the **correct place** in the sorted order (but the other doors may not be).

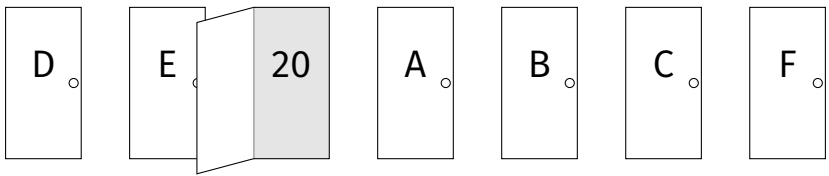
But, every door to the left is smaller (≤), every door to the right is larger (≥).

In general...

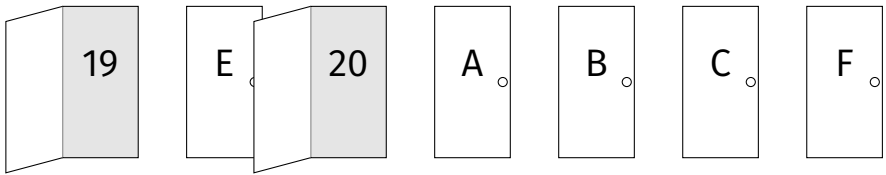
- Let's generalize strategy for kth order statistic.
- Example: $k = 2$.

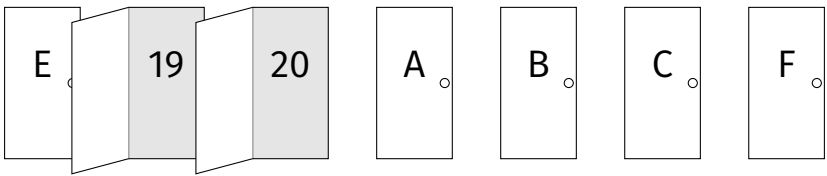

A
B
C
D
E
F
G


A
B
C
D
E
F
20



after partitioning


19
E
20
A
B
C
F



after partitioning

Strategy

- Open arbitrary door (that hasn't been ruled out).
- **Partition** the doors around this number:
  - Move doors smaller than this to the left,
  - Larger than this to the right.
- Let p be our door's new position, k be the order we want.
  - If $p = k$, return this door.
  - If $p < k$, rule out doors to left.
  - If $p > k$, rule out doors to right.
- Repeat.

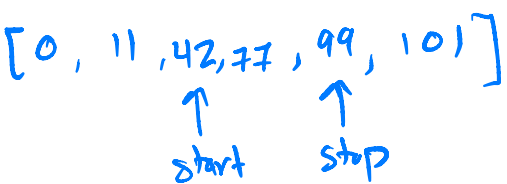
In Code

```
import random
def quickselect(arr, k, start, stop):
    """Finds kth order statistic in numbers[start:stop])"""
    pivot_ix = random.randrange(start, stop)
    pivot_ix = partition(arr, start, stop, pivot_ix)
    pivot_order = pivot_ix + 1
    if pivot_order == k:
        return arr[pivot_ix]
    elif pivot_order < k:
        return quickselect(arr, k, pivot_ix + 1, stop)
    else:
        return quickselect(arr, k, start, pivot_ix)
```

Example



`arr = [77, 42, 11, 99, 0, 101]` `k = 3`



DSC 40B

Theoretical Foundations II

Lecture 7 | Part 5

Partition

Partitioning

- Given an array of n numbers and the index of a **pivot** p.
- Rearrange elements so that:
  - Everything $< p$ is first.
  - Everything $= p$ is next.
  - Everything $> p$ is last.
- Return index of first element $\geq p$.

```
def partition(arr, start, stop, pivot_ix):
    """Partition arr[start:stop] around pivot."""
    left = []
    pivot_count = 0
    right = []
    pivot = arr[pivot_ix]
    for ix in range(start, stop):
        if arr[ix] < pivot:
            left.append(arr[ix])
        elif arr[ix] == pivot:
            pivot_count += 1
        else:
            right.append(arr[ix])
    ix = start
    for x in left:
        arr[ix] = x
        ix += 1
    for i in range(pivot_count):
        arr[ix] = pivot
        ix += 1
    for x in right:
        arr[ix] = x
        ix += 1
    return start + len(left)
```

Partition

- `partition` takes $\Theta(n)$ time.
  - This is **optimal**.
- But we can use memory more efficiently.

Motivation

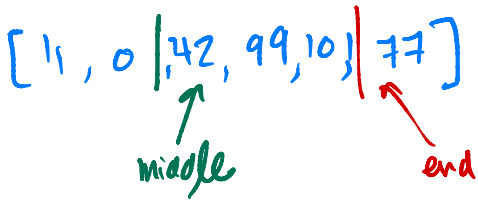
- Similar to selection sort, we'll use **two** barriers:
- **"Middle"** barrier:
  - Separates things < pivot from things ≥
  - Index of first thing in "right"
- **"End"** barrier:
  - Separates processed from processed.
  - Index of first "unprocessed" thing.



Example

Simplification: start by moving pivot to end.

```
arr = [77, 42, 11, 99, 0, 101]    pivot_ix = 1
```



Loop Invariants

- After each iteration:
  - everything in `arr[start:middle_barrier]` is < pivot.
  - everything in `arr[middle_barrier:end_barrier]` is ≥ pivot.
  - everything in `arr[end_barrier:stop]` is “unprocessed”

```
def in_place_partition(arr, start, stop, pivot_ix):
    def swap(ix_1, ix_2):
        arr[ix_1], arr[ix_2] = arr[ix_2], arr[ix_1]

    pivot = arr[pivot_ix]
    swap(pivot_ix, stop-1)
    middle_barrier = start
    for end_barrier in range(start, stop - 1):
        if arr[end_barrier] < pivot:
            swap(middle_barrier, end_barrier)
            middle_barrier += 1
        # else:
            # do nothing
    swap(middle_barrier, stop-1)
    return middle_barrier
```

$\Theta(n)$

$n = stop - start$

Efficiency

- Also takes $\Theta(n)$ time.
- No auxiliary memory required.

DSC 40B

Theoretical Foundations II

Lecture 7 | Part 6

Time Complexity Analysis

Time Complexity

- What is time complexity of `quickselect`?

$T(n) = \Theta(n) + T(?)$

```
import random
def quickselect(arr, k, start, stop):
    """Finds kth order statistic in numbers[start:stop])"""
    pivot_ix = random.randrange(start, stop)
    pivot_ix = partition(arr, start, stop, pivot_ix)
    pivot_order = pivot_ix + 1
    if pivot_order == k:
        return arr[pivot_ix]
    elif pivot_order < k:
        return quickselect(arr, k, pivot_ix + 1, stop)
    else:
        return quickselect(arr, k, start, pivot_ix)
```

Problem

- We don’t know the size of the subproblem.
  - Is random, can be anywhere from 1 to $n - 1$.
- Difficult to write recurrence relation.

Good and Bad Pivots

- Some pivots are better than others.
  - **Good**: splits array into roughly balanced halves.
  - **Bad**: splits array into wildly unbalanced pieces.

Exercise

Suppose we're searching for the minimum. What would be the worst possible pivot?

Worst Case

- Suppose we're searching for $k = 1$ (minimum).
- Worst pivot: the maximum.
- Worst case: use max as pivot every time.
- Subproblem size: $n - 1$.

Worst Case

- Every recursive call is on problem of size $n - 1$.
- $T(n) = T(n - 1) + \Theta(n)$.
  - Solution: $\Theta(n^2)$.
- Intuitively, randomly choosing largest number as pivot every time is **very** unlikely!

$$\frac{1}{n} \times \frac{1}{n-1} \times \frac{1}{n-2} \times \cdots \times \frac{1}{3} \times \frac{1}{2} = \frac{1}{n!}$$

Equally Unlikely

- Pivot falls exactly in the middle, every time.
- Subproblems are of size $n/2$.
- $T(n) = T(n/2) + \Theta(n)$.
  - Solution: $\Theta(n)$.

Typically

- Pivot falls somewhere in the middle.
- Sometimes **good**, sometimes **bad**.
- But **good** pivots reduce problem size by **so much** that they make up for **bad** pivots.

Analogy

- You’re 100 miles away from home.
- You have a button that, if you press it, teleports you **1 mile** closer to home.
- How many times must you press it before you’re 1 mile away from home?

Analogy

- You're 100 miles away from home.
- You have a button that, if you press it, teleports you **1 mile** closer to home.
- How many times must you press it before you're 1 mile away from home?
  - **Answer**: 99 times.

Analogy

- You're 100 miles away from home.
- You have a button that, if you press it, teleports you **half the distance** to home.
- How many times must you press it before you're <1 mile away from home?

Analogy

- You're 100 miles away from home.
- You have a button that, if you press it, teleports you **half the distance** to home.
- How many times must you press it before you're <1 mile away from home?
  - **Answer**: *about* $\log_2 100 \approx 6.64$ times.

Analogy

- You're 100 miles away from home.
- You have a button that, if you press it, teleports you **half the distance** to home with probability 1/2, does nothing with probability 1/2.
- How many times do you **expect** to press it before you're <1 mile away from home?
  - **Answer**: *about* twice as many times as before. So, $2 \log_2 100 \approx 13.28$.

Quickselect

- The same reasoning applies to quickselect.
- If we always get a **good** pivot, time taken is $\Theta(n)$.
- If half the time we get a **bad** pivot, we expect:
  - To make twice as many recursive calls.
  - Take twice as much time as before.
- But $2\Theta(n) = \Theta(n)$.

Quickselect

- Expected time complexity: $\Theta(n)$.
- Worst case: $\Theta(n^2)$, but **very unlikely**.

Median

- We can find the median in expected linear time with **quickselect**.

DSC 40B

Theoretical Foundations II

Lecture 7 | Part 7

Quicksort

Last Time

- We saw `mergesort`.
- **Divide**: split list directly down the middle
- **Conquer**: sort each half
- **Combine**: merge sorted halves together

`merge` does all the work

- In `mergesort`, we are lazy when we divide.
- So we have to work to combine.

`[4,1,3,2]` → `[4,1]`,`[3,2]` → `[4,3]`,`[2,3]` → `[1,2,3,4]`

What if?

- Suppose we divide so that everything in `left` is smaller than everything in `right`:
- After sorting, no need for `merge`.
- `[5,1,3,8,6,2]` → `[1,3,2]`,`[5,8,6]`

What if?

- Suppose we divide so that everything in `left` is smaller than everything in `right`:
- After sorting, no need for `merge`.
- `[5,1,3,8,6,2]` → `[1,3,2]`,`[5,8,6]`
- This is what `partition` does!

Quicksort

```
def quicksort(arr, start, stop):
    """Sort arr[start:stop] in-place."""
    if stop - start > 1:
        pivot_ix = random.randrange(start, stop)
        pivot_ix = partition(arr, start, stop, pivot_ix)
        quicksort(arr, start, pivot_ix)
        quicksort(arr, pivot_ix+1, stop)
```

Time Complexity

- Average case: $\Theta(n \log n)$
- Worst case: $\Theta(n^2)$.
- Like with `quickselect`, worst case is **very rare**.

Mergesort vs Quicksort

- Mergesort has better worst case complexity.
- But in practice, Quicksort is often faster.
- Takes less memory, too.

Memory Requirements

- `merges` requires output array, $\Theta(n)$ additional space.
- `partition` works in-place, requires no additional space[3]
- Example: sorting 3 GB of data with 4 GB of RAM.

[3] Call stack for `quicksort` requires $\Theta(\log n)$ additional space.